

ABOUT

ENGINEER WITH 10+ YEARS EXPERIENCE ACROSS AUTOMATION, MACHINE-SOFTWARE, ROBOTICS, MANUFACTURING, AND EMBEDDED SYSTEMS, DELIVERING PRODUCTION GRADE SOLUTIONS IN COMPLEX INDUSTRIAL ENVIRONMENTS. I BRING SYSTEM LEVEL THINKING AND TAKE OWNERSHIP OF DESIGNING, BUILDING, AND ARCHITECTING IMPACTFUL ENGINEERING SOLUTIONS.

EDUCATION

MS, COMPUTER ENGINEERING (3.41/4) – DEC-17 

NEW YORK UNIVERSITY 

BE, INSTRUMENTATION & CONTROLS ENGINEERING (7.04/10) – JUN-14 

GUJARAT TECHNOLOGICAL UNIVERSITY

SKILLS

COMPUTER PROGRAMMING:	C/C++, CS, JAVA, PYTHON (AI-AUTOMATION TOOLS, LIMITED HANDS-ON CODING), WEB(HTML, CSS, JS)
AUTOMATION:	PLCS(LADDER-LOGIC, FBD), HMI/SCADA MICRO-CONTROLLERS & SINGLE BOARD COMPUTERS
SCM:	GIT (BASH & SOURCETREE)
INDUSTRIES SERVED:	MACHINERIES, MANUFACTURING, CONTROL PANELS, SOLAR ENERGY, EDUCATION

PROFESSIONAL EXPERIENCE

UNIVERSAL INSTRUMENTS, NY

 SOFTWARE ENGINEER III - AUTOMATION

Sep 22 - present

Core Responsibilities:

- Maintain and enhance high-precision, high-speed software modules for Fuzion SMT Platform.
- Provide remote production-line support by investigating issues, analyzing logs, replicating site conditions to perform root-cause analysis, reviewing configurations and delivering software fixes, patches, and release updates.
- Enable field and application engineers to correctly set up and validate applications by offering technical guidance, interpreting machine data and supplying required configuration or software updates.

Key contributions:

- Built the SBS feature integrating Keyence SR-X300, delivering Fuzion's first side-barcode scanning capability with live-view, traceability, and error-recovery support.
- Upgraded the AMS workflow from one point to 13 locations on wafer, delivering a major accuracy improvement in wafer placement and calibration.
- Built the wafer-to-board mapping system that converts customer wafer maps into Fuzion board templates, enabling direct use of wafer data and eliminating hours of manual template creation.
- Delivered Selective Placement, enabling Fuzion's first smart pick/buffer/place workflow using DFF-based categories and avoiding-priority rules, improving overall product quality.
- Implemented AVP, adding user-defined fiducial capability beyond fixed shapes and improving alignment flexibility across ECS, NPI, and Production workflows.
- Integrated Cognex QuickBuild (VPP) with MCOS↔QBIT TCP/IP triggers and full reject/retry handling, enabling Dispense Quality Inspection on the Fuzion Precision-Bonder platform for the first time.
- Delivered key platform enhancements — Light Intensity Controller, nested-tray support, isosceles fiducial, and configurable park-position — among many other features across Fuzion.
- Tech-stack : · CS · C · CPP · Git · VxWorks · Sourcetree.

NOR-CAL CONTROLS, CA

☐ AUTOMATION ENGINEER

Jan 22 - Jul 22

- Developed automation software for utility-scale solar power plants.
- Assisted commissioning and troubleshooting of PLC/SCADA systems.
- Tech stack: Ignition SCADA · GE PLC · SEL-RTAC · Red Lion

MINI-CIRCUITS, NY

☐ SOFTWARE ENGINEER - MACHINE AUTOMATION

Feb 18 - Nov 21

- Developed and maintained automation software for rotary machines with computer-vision inspection.
- Implemented reverse-motion support on rotary machines with a pressing arm, enabling EVT testing.
- Delivered the new shopping-cart system for online store, improving performance and customer experience.
- Built enterprise software used across internal teams to streamline workflows reducing manual efforts.
- Tech stack: Java · JavaFX · HTML · CSS · JavaScript.

PRIMA AUTOMATION, AHMEDABAD, INDIA

☐ DESIGN ENGINEER

Jun 15 - Jan 16

- Programmed Siemens PLCs in TIA Portal and developed HMI interfaces for industrial automation.
- Designed electrical and control panels using AutoCAD.
- Created detailed electrical schematics and wiring diagrams using E-plan.

☐ QUALITY CONTROL ENGINEER

Jun 14 - Jun 15

- Electrical power testing, continuity checks and full Control Panel functional verification.
- BOM verification, component & ferrule naming checked against engineering specs.
- Inspected wiring for looseness, verified terminations, and identified any physical or assembly damage.
- PLC/HMI/SCADA functional testing to confirm full panel operation before dispatch.

ACADEMIC WORK EXPERIENCE

REVMAX FLEET OPTIMIZATION, NY

☐ DEVELOPER INTERN

Sep - Dec 17

- Engineered Doppler-radar firmware for the SidewalkIQ pedestrian-monitoring sensor.
- Supported deployment and tested sensors across multiple NYC to validate field performance.
- Improved data consistency and optimized backend storage/transfer pipelines.
- Tech stack : · Arduino-C ·

NEW YORK UNIVERSITY, NY

☐ INSTRUCTOR (ITEST PROGRAM), ROBOTICS & MECHATRONICS DPT.

Sep - Dec 17

- Mentored Robotics Club programmers for FIRST Robotics and VEX competitions.
- Taught foundational programming concepts to student robotics teams.

☐ LEAD INSTRUCTOR (CREST PROGRAM), CENTER FOR K12 STEM EDUCATION

May - Aug 17

- Mentored high-school interns in Arduino, IoT, robotics, and 3D printing & led STEM projects including:
 - Autonomous Car (Mouse in a Maze) - STM32 Board: https://github.com/sap586/RTES_Project
 - Wireless Robot Controller - Arduino: <https://github.com/sap586/WirelessRobotDay5Activity>

NEOPLAST (I) PVT. LTD., AHMEDABAD, INDIA

☐ ENGINEERING TRAINEE

Aug 13 - May 14

- Created a dummy working-principal based demo for 'PVC Compounding & Material handling system' using Siemens S700 and TIA-Portal.

This document is prepared using L^AT_EX. Click [here](#) to download the source code.